

Template Service DB design and Access Patterns

Access Patterns

Write Patterns

Create / Update Template Version

Operation	Record Type	PK	SK	Action
1	Template Version Snsnot	TPL# <TemplateId>	VER# <Version>	Put
2	Latest Pointer	TPL# <TemplateId>	LATEST	Update
3	Audit Record	TPL# <TemplateId>	AUDIT# <Timestamp>	Put

Publish Template Version

Operation	Record Type	PK	SK	Action
1	Latest Pointer	TPL# <TemplateId>	LATEST	Update
2	Activity Record	TPL# <TemplateId>	ACT# <Timestamp>	Put
3	Audit Record	TPL# <TemplateId>	AUDIT# <Timestamp>	Put

Create Org Derived Template

Operation	Record Type	PK	SK	Action
1	Org Template Version	ORG# <OrgId>#TPL# <TemplateId>	VER# <Version>	Put

2	Org Latest Pointer	ORG# <OrgId>#TPL# <TemplateId>	LATEST	Update
3	Lineage Record	TPL# <MasterTemplateId>	LINEAGE# <OrgId># <Version>	Put
4	Audit Record	ORG# <OrgId>#TPL# <TemplateId>	AUDIT# <Timestamp>	Put

Create Org Enablement

Operation	Record Type	PK	SK	Action
1	Org Enablement	ORG#<OrgId>	ENABLE# <MasterTemplateVersionId>	Put
2	Audit Record	ORG#<OrgId>	AUDIT# <Timestamp>	Put

Upgrade Org Template

Operation	Record Type	PK	SK	Action
1	Org Template Version	ORG# <OrgId>#TPL# <TemplateId>	VER# <NewVersion>	Put
2	Org Latest Pointer	ORG# <OrgId>#TPL# <TemplateId>	LATEST	Update
3	Upgrade Lineage	TPL# <TemplateId>	LINEAGE#UPGRADE#<Version>	Put
4	Audit Record	ORG# <OrgId>#TPL#	AUDIT# <Timestamp>	Put

		<TemplateId>		
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Read Patterns

1 : Get Template by TemplateVersionID

Operation	Record Type	PK	SK	Action
1	Template Version	TPL# <TemplateId>	VER# <Version>	GetItem

2 : Get Latest Published Template for Profile

Operation	Record Type	PK	SK	Action
1	Latest Pointer	TPL# <TemplateId>	LATEST	GetItem
2	Template Version	TPL# <TemplateId>	VER# <latestPublishedVersion>	GetItem

3 : Get All Versions of Template

Operation	Record Type	PK	SK	Action
1	Template Versions	TPL# <TemplateId>	begins_with(VER#)	Query

4 : List Templates by Type/Category/Condition/Country

Operation	Record Type	GSI PK	GSI SK	Action
1	Template Version	TYPE# <TemplateType>	CAT# <Category>#COND# <Condition>#COUNTRY#	Query GSI1

			<Country>#STA TUS#<Status>	
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5 : Get All Published Templates

Operation	Record Type	GSI PK	GSI SK	Action
1	Published Template	TYPE# <TemplateType >	STATUS#PUBLIS HED	Query GSI1

6 : Get Org Enabled Templates

Operation	Record Type	PK	SK	Action
1	Org Enablement	ORG#<OrgId>	ENABLE# <MasterTempla teVersionId>	Query

7 : Get Org Derived Templates

Operation	Record Type	PK	SK	Action
1	Org Template	ORG#<OrgId>	TPL# <TemplateVers ionId>	Query

8 : Get Draft Templates Pending Review

Operation	Record Type	GSI PK	GSI SK	Action
1	Draft Template	TYPE# <TemplateType >	STATUS#DRAFT	Query GSI1

9 : Compare Two Template Versions

Operation	Record Type	PK	SK	Action
1	Template Version A	TPL# <TemplateId>	VER# <VersionA>	BatchGet
2	Template Version B	TPL# <TemplateId>	VER# <VersionB>	BatchGet

10 : Get Audit History for Template

Operation	Record Type	PK	SK	Action
1	Audit Records	TPL# <TemplateId>	begins_with(A UDIT#)	Query

11 : Get Activity Timeline

Operation	Record Type	PK	SK	Action
1	Activity Records	TPL# <TemplateId>	begins_with(A CT#)	Query

12 : Get Templates by Status

Operation	Record Type	GSI PK	GSI SK	Action
1	Template Version	TYPE# <TemplateType >	STATUS# <Status>	Query GSI1

13 : Get Latest Org Template Version

Operation	Record Type	PK	SK	Action
1	Org Latest Pointer	ORG# <OrgId>#TPL# <TemplateId>	LATEST	GetItem

2	Org Template Version	ORG# <OrgId>#TPL# <TemplateId>	VER# <activeVersion>	GetItem
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14 : Get All Enablements for Org

Operation	Record Type	PK	SK	Action
1	Org Enablement	ORG#<OrgId>	begins_with(ENABLE#)	Query

15 : Soft Delete Template

Operation	Record Type	PK	SK	Action
1	Template Version	TPL# <TemplateId>	VER# <Version>	Update
2	Audit Record	TPL# <TemplateId>	AUDIT# <Timestamp>	Put

16 : Search Templates by Name

Operation	Record Type	GSI PK	GSI SK	Action
1	Search Projection	TYPE# <TemplateType>	NAME# <TemplateName>	Query GSI1

18 : Get All Templates Linked to Master Version

Operation	Record Type	GSI PK	GSI SK	Action
1	Org Derived Template	MASTER# <MasterTemplateVersionId>	ORG# <OrgId>#TPL# <TemplateVersionId>	Query GSI3

19 : Get All Orgs Using Master Template

Operation	Record Type	GSI PK	GSI SK	Action
1	Org Enablement	MASTER# <MasterTemplateVersionId>	ORG#<OrgId>	Query GSI3

20 : Get Published Template for Exact Profile

Operation	Record Type	PK	SK	Action
1	Latest Pointer	TPL# <TemplateId>	LATEST	GetItem
2	Published Template Version	TPL# <TemplateId>	VER# <latestPublishedVersion>	GetItem

#	Access Pattern	Query Type	Frequency	Notes
1	Get template by TemplateVersionID	PK + SK	Very High	Core read path
2	Get latest published template for profile	PK + LSI	Very High	Publish/runtime usage
3	Get all versions of a template	PK query	High	Version history
4	List templates by type/category/condition/country	GSI	High	Admin listing
5	Get all published templates	GSI	Medium	Selection/linking
6	Get org enabled templates	GSI	High	Org onboarding

7	Get org-derived templates	GSI	High	Org management
8	Get draft templates pending review	GSI	Medium	Review workflow
9	Compare two template versions	Batch Get	Medium	Upgrade workflow
10	Get audit history for template	PK query	Medium	Audit/governance
11	Get activity timeline	PK query	Medium	Timeline UI
12	Get templates by status	GSI	Medium	Filtering
13	Get latest org template version	PK + LSI	High	Runtime resolution
14	Get all enablements for org	GSI	Medium	Org governance
15	Soft delete template	Update	Low	Status update only
16	Get template lineage	PK query	Medium	Derivation/version tracing
17	Search templates by name	GSI	Medium	UI search
18	Get all templates linked to master version	GSI	Medium	Upgrade impact
19	Get all orgs using master template	GSI	Medium	Governance/reporting
20	Get published template for exact profile	PK + LSI	Very High	Critical runtime path

Primary Key Design

Attribute	Purpose
PK	Entity grouping
SK	Entity hierarchy/versioning

Entity Structure

Entity	PK	SK
Template Root	TPL#<TemplateId>	META
Template Version	TPL#<TemplateId>	VER#<Version>
Audit Record	TPL#<TemplateId>	AUDIT# <Timestamp>
Activity Record	TPL#<TemplateId>	ACT#<Timestamp>
Org Enablement	ORG#<OrgId>	ENABLE# <MasterTemplateVersionId>
Org Derived Template	ORG#<OrgId>	TPL# <TemplateVersionId>

Attributes

Attribute	Example
templateId	CP-HTN-001
templateVersionId	CP-HTN-001-V01
templateType	CARE_PLAN
category	CHRONIC
condition	HTN
country	US
status	PUBLISHED

version	1
ownerOrgId	ORG-001
masterTemplateVersionId	CP-HTN-001-V01
latestPublished	true
deleted	false

LSI Design

LSI	Sort Key
LSI1	status
LSI2	version
LSI3	lastModifiedDate
LSI4	publishedDate
LSI5	ownerOrgId#status

GSI Design

GSI	GSI PK	GSI SK	Access Patterns Supported	Purpose
GSI1 : Template Discovery	TYPE# <Template Type>	CAT# <Category >#COND# <Conditio n>#COUNTR Y# <Country> #STATUS# <Status>	4, 5, 12, 17, 20	List/search/fi lter templates by type, category, condition, country, and status
GSI2 : Org Governance	ORG# <OrgId>	STATUS# <Status># TPL#	6, 7, 14	Get org enabled templates,

		<Template Type>		org-derived templates, org governance views
GS13 : Master Lineage	MASTER# <MasterTe mplateVer sionId>	ORG# <OrgId>#T PL# <Template VersionId >	18, 19	Find all org templates derived from a master version and all orgs using a master template

